UNIVERSITY OF CAMBRIDGE | Yusuf Hamied Department of Chemistry

Presentation title: Effective Radiative Forcing from Very Short-Lived Halocarbons from UKESM1 with Fully Interactive Halocarbon Chemistry

Historical Halocarbon Emissions Enhance OH and H_2O_2 Production leading to Cloud Formation

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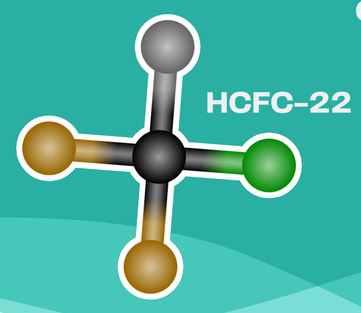
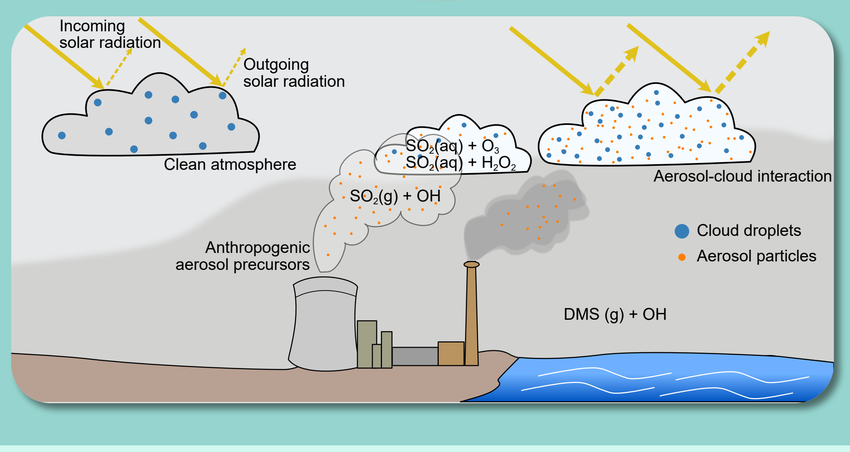


Background



- Ozone-depleting substances, such as **halocarbons**, cause a **reduction in stratospheric ozone**
- **Ozone and OH production in the troposphere increases** as a response through ultraviolet radiation reaching the troposphere
- Among many interactions, **OH is involved in sulfate aerosol formation**
- Indirect radiative effects and climate adjustments have *low confidence* according to IPCC AR6

Results

1. Stratospheric O_3 depletion leads to increased tropospheric OH formation

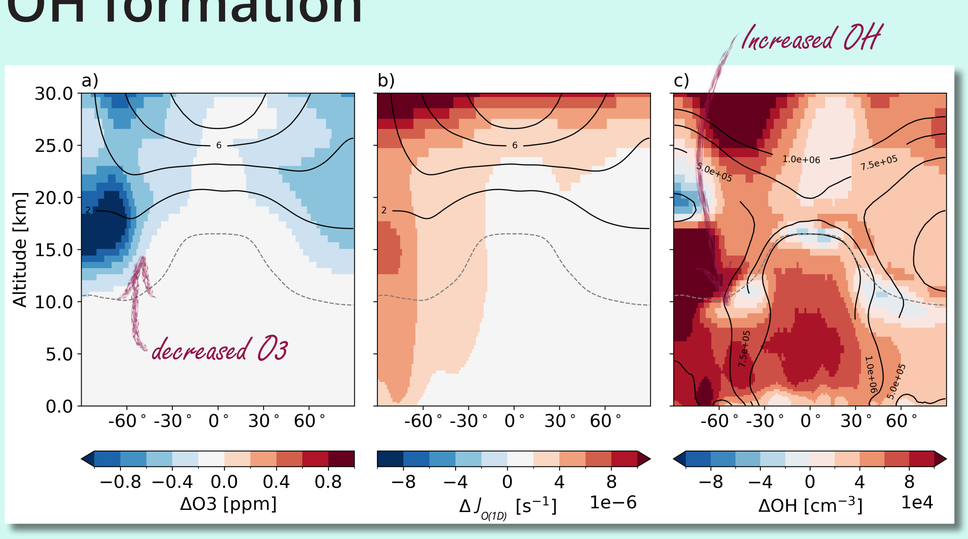


Figure 1:

a) Zonal mean changes in O_3 mixing ratio due to present-day halocarbons. Contour shows O_3 in pre-industrial simulation.

b) Changes in photolysis rate of O_3 to $O(^1D)$

$$O_3 + h\nu \rightarrow O(^1D) + O_2$$

c) Changes to OH concentration. Contour shows OH in pre-industrial simulation.

$$O(^1D) + H_2O \rightarrow 2\ OH$$

2. OH and H_2O_2 increase and react with SO_2, forming sulfate aerosol over Southern Hemisphere

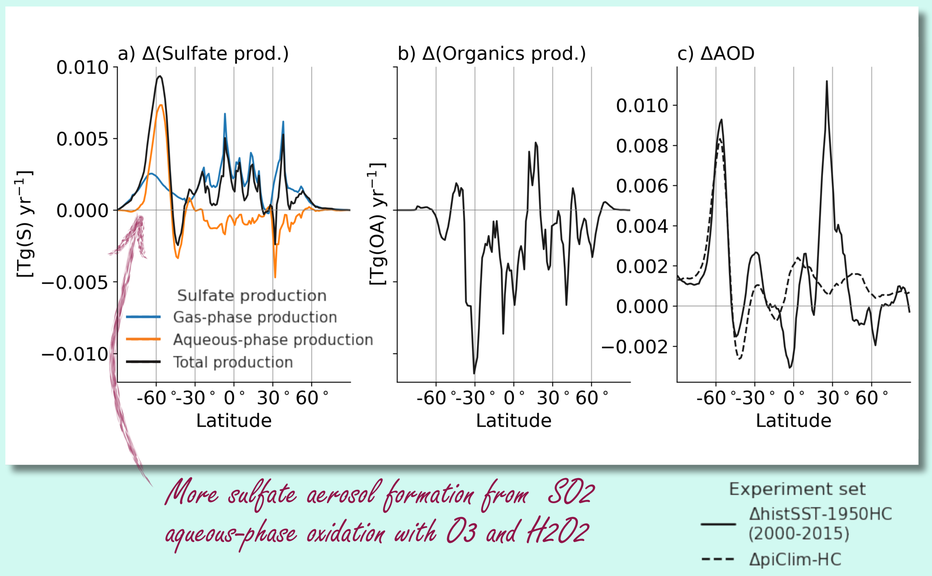


Figure 2:

a) Changes in sulfate aerosol production in gas- and aqueous-phase

b) Reduction in organic aerosol secondary production from BVOCs

c) Increases in AOD in the Southern Hemisphere

3. More aerosol loading promotes cloud formation, leading to negative ERF

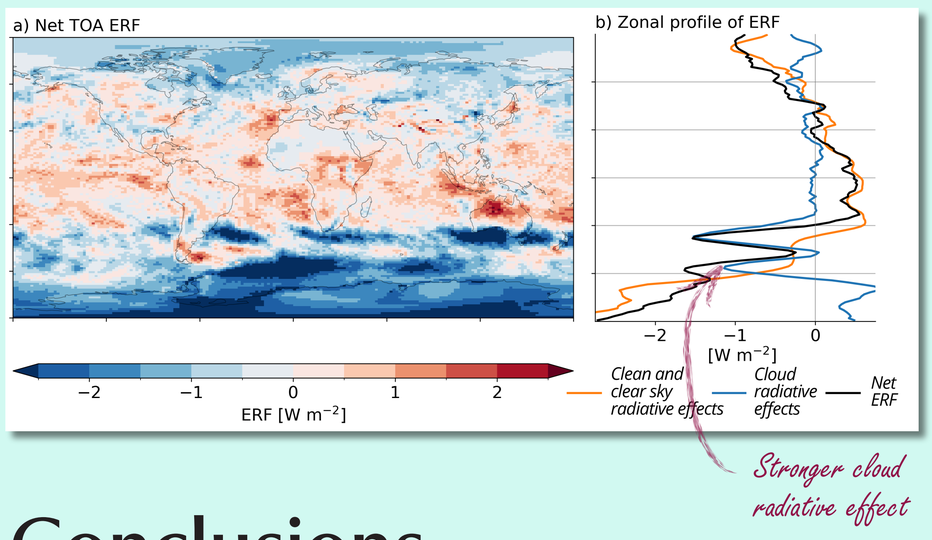


Figure 2:

a) Global distribution of net top-of-atmosphere effective radiative forcing (ERF)

b) Zonal profile of ERF components

ERF = -0.18 W m^{-2}
ERFcsaf = 0.2 W m^{-2}
IRF= -0.02 W m^{-2}

Simulation design

AerChemMIP | CMIP6

- Existing UKESM1 outputs from the AerChemMIP were used in this work.
- All experiments have prescribed sea-surface temperature (SST). All results are from comparing *piClim-control* and *piClim-HC* which prescribe lower boundary condition for CFC-11 and CFC-12 and CH_3Br at the 2014 level
- 45 simulation years, using the final 35 year for analysis

Conclusions

- We have established that CFC-11 and CFC-12 induce chemistry-aerosol-climate feedback through changes in tropospheric OH
- The standard UKESM1 has limited halocarbon chemistry mechanism.
- We are updating the extended and interactive halocarbon chemistry to investigate the changes from other short-lived substances such as HCFCs

References

Voulgarakis A., *et al*. ACP. (2013); Szopa S., et al. (2021). IPCC WG1; Collins, W. J., *et al*. (2026). EGUsphere [preprint]

Further Information

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